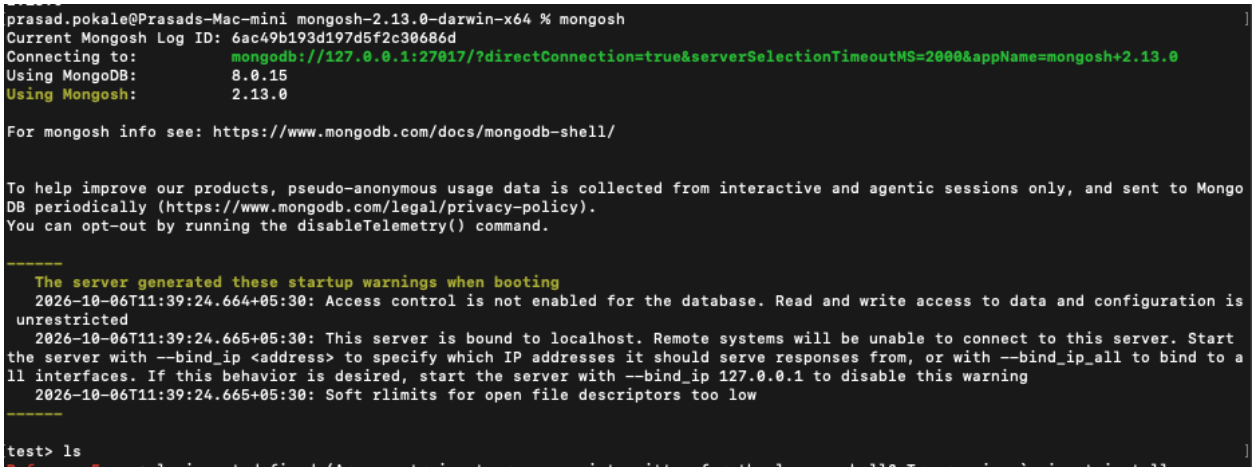
Assignment Mongodb Basic

Task 1 : Local Setup and Data Creation

Step 1 : Verify MongoDB

Command : mongosh



Command : show dbs



Step 2 : Select the Database



Step 3: Insert the Dataset

Command : db.employees.insertMany([....])

```
company> db.employees.insertMany([
|   {
|     _id: "EMP001",
|     name: "John",
|     department: "Engineering",
|     experience: 4,
|     skills: ["Java", "SpringBoot"],
|     active: true,
|     address: {
|       city: "Pune",
|       country: "India"
|     }
|   },
|   {
|     _id: "EMP002",
|     name: "Alice",
|     department: "HR",
|     experience: 3,
|     skills: ["Recruitment", "Communication"],
|     active: true,
|     address: {
|       city: "Mumbai",
|       country: "India"
|     }
|   },
|   {
|     _id: "EMP003",
|     name: "David",
|     department: "Engineering",
|     experience: 6,
|     skills: ["Java", "MongoDB"],
|     active: true,
|     address: {
|       city: "Bengaluru",
|       country: "India"
|     }
|   },
|   {
|     _id: "EMP004",
|     name: "Emma",
|     department: "Finance",
|     experience: 2,
|     skills: ["Accounting", "Excel"],
|     active: false,
|     address: {
|       city: "Pune",
|       country: "India"
|     }
|   },
|   {
|     _id: "EMP005",
|     name: "Robert",
|     department: "Engineering",
|     experience: 5,
|     skills: ["Java", "Docker"],
|     active: true,
|     address: {
|       city: "Delhi",
|       country: "India"
|     }
|   }
[ ])
{
  acknowledged: true,
  insertedIds: {
    '0': 'EMP001',
```

Verify :
Company Database exists :
Command : show dbs

```
company> show dbs
admin     40.00 KiB
company   40.00 KiB
config   108.00 KiB
local     40.00 KiB
```

The employees collection exists.
Command : show collections

```
company> show collections
employees
```

The collection contains 5 documents.
Command: db.employees.countDocuments()

```
company> db.employees.countDocuments()
5
```

Task 2: Read and Query Operations

2.1 Engineering Employees

Command :

```
db.employees.find(
  { department: "Engineering" },
  { _id: 1, name: 1, experience: 1 }
).sort({ name: 1 })
```

Output :

```
company> db.employees.find(
|   { department: "Engineering" },
|   { _id: 1, name: 1, experience: 1 }
| ).sort({ name: 1 })
[
  { _id: 'EMP003', name: 'David', experience: 6 },
  { _id: 'EMP001', name: 'John', experience: 4 },
  { _id: 'EMP005', name: 'Robert', experience: 5 }
]
```

2.2 Employees with 5 or More Years of Experience

Command :

```
db.employees.find(
  { experience: { $gte: 5 } },
  { _id: 0, name: 1, experience: 1 }
).sort({ experience: -1 })
```

Output :

```
company> db.employees.find(
|   { experience: { $gte: 5 } },
|   { _id: 0, name: 1, experience: 1 }
| ).sort({ experience: -1 })
[ { name: 'David', experience: 6 }, { name: 'Robert', experience: 5 } ]
```

2.3 Nested Document Query

Command :

```
db.employees.find(
  { "address.city": "Pune" },
  { _id: 0, name: 1, "address.city": 1 }
).sort({ name: 1 })
```

Output :

```
company> db.employees.find(
|   { "address.city": "Pune" },
|   { _id: 0, name: 1, "address.city": 1 }
| ).sort({ name: 1 })
[
  { name: 'Emma', address: { city: 'Pune' } },
  { name: 'John', address: { city: 'Pune' } }
]
```

2.4 Array Query

Command :

```
db.employees.find(
  { skills: { $in: ["MongoDB", "Spring Boot"] } },
  { _id: 0, name: 1, skills: 1 }
).sort({ name: 1 })
```

Output :

```
company> db.employees.find(
|   { skills: { $in: ["MongoDB", "Spring Boot"] } },
|   { _id: 0, name: 1, skills: 1 }
| ).sort({ name: 1 })
[ { name: 'David', skills: [ 'Java', 'MongoDB' ] } ]
```

2.5 Sorting and Limiting
Command :

```
db.employees.find(
  {},
  { _id: 0, name: 1, experience: 1 }
).sort({ experience: -1 }).limit(2)
```

Output :

```
company> db.employees.find(
|   {},
|   { _id: 0, name: 1, experience: 1 }
| ).sort({ experience: -1 }).limit(2)
[ { name: 'David', experience: 6 }, { name: 'Robert', experience: 5 } ]
```

Task 3: Update Operations

3.1 $inc : Increase Alice's experience by 1.

Command :

```
db.employees.updateOne(
  { name: "Alice" },
  { $inc: { experience: 1 } }
)
```

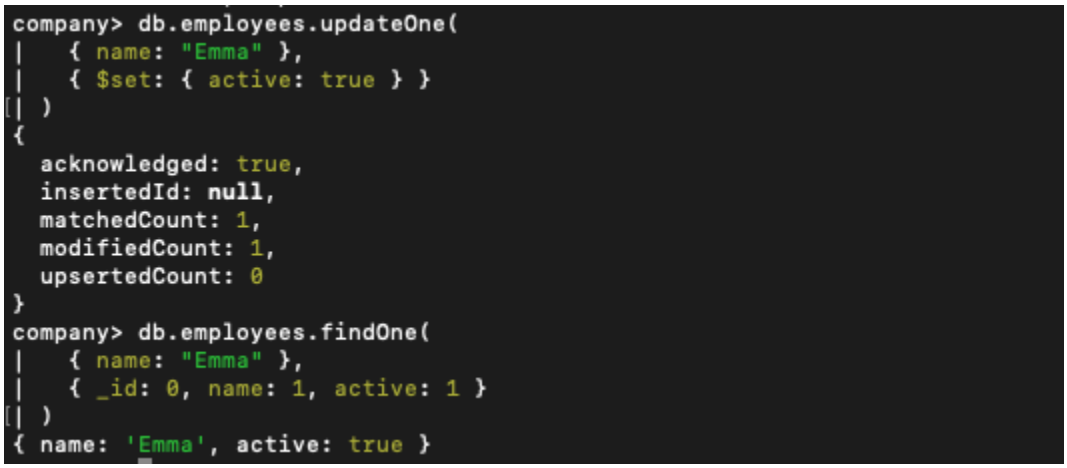
Output :

```
company> db.employees.updateOne(
|   { name: "Alice" },
|   { $inc: { experience: 1 } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
company> db.employees.findOne(
|   { name: "Alice" },
|   { _id: 0, name: 1, experience: 1 }
| )
{ name: 'Alice', experience: 4 }
```

3.2 $set: Change Emma's active value from: false to true
Command :

```
db.employees.updateOne(
  { name: "Emma" },
  { $set: { active: true } }
)
```

Output :



3.3 $addToSet: Add MongoDB to Robert's skills array using $addToSet.

Command :

```
db.employees.updateOne(
  { name: "Robert" },
  { $addToSet: { skills: "MongoDB" } }
)
```

Output :

```
company> db.employees.updateOne(
|   { name: "Robert" },
|   { $addToSet: { skills: "MongoDB" } }
[| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
company> db.employees.findOne(
|   { name: "Robert" },
|   { _id: 0, name: 1, skills: 1 }
[| )
{ name: 'Robert', skills: [ 'Java', 'Docker', 'MongoDB' ] }
```

Task 4: Delete Operation

```
company> db.employees.updateOne(
|   { name: "Robert" },
|   { $addToSet: { skills: "MongoDB" } }
[| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
company> db.employees.findOne(
|   { name: "Robert" },
|   { _id: 0, name: 1, skills: 1 }
[| )
{ name: 'Robert', skills: [ 'Java', 'Docker', 'MongoDB' ] }
company> db.employees.insertOne({
|   _id: "EMP999",
|   name: "Temporary Employee",
|   department: "Training",
|   experience: 0
[| })
{ acknowledged: true, insertedId: 'EMP999' }
company> db.employees.deleteOne({
|   _id: "EMP999"
[| })
{ acknowledged: true, deletedCount: 1 }
company> db.employees.findOne({
|   _id: "EMP999"
[| })
null
```

Task 5: Indexing

5.1 Check Existing Indexes

Command :

```
db.employees.getIndexes()
```

Output :

```
company> db.employees.getIndexes()
[ { v: 2, key: { _id: 1 }, name: '_id_' } ]
```

5.2 Create an Index

Command :

```
db.employees.createIndex({ department: 1 })
```

Output :

```
company> db.employees.createIndex({ department: 1 })
department_1
```

5.3 Verify the Index

Command :

```
db.employees.getIndexes()
```

Output :

```
company> db.employees.getIndexes()
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { department: 1 }, name: 'department_1' }
]
```

Task 6: Aggregation

6.1 Employees per Department

Command :

```
db.employees.aggregate([
 {
  $group: {
   _id: "$department",
   count: { $sum: 1 }
  }
 },
 {
  $sort: {
   _id: 1
  }
 }
])
```

Output :

```
company> db.employees.aggregate([
|   {
|     $group: {
|       _id: "$department",
|       count: { $sum: 1 }
|     }
|   },
|   {
|     $sort: {
|       _id: 1
|     }
|   }
| ])
[
  { _id: 'Engineering', count: 3 },
  { _id: 'Finance', count: 1 },
  { _id: 'HR', count: 1 }
]
```

6.2 Average Experience per Department
Command :

```
db.employees.aggregate([
  {
    $group: {
      _id: "$department",
      averageExperience: { avg: "experience" }
    }
  },
  {
    $sort: {
      _id: 1
    }
  }
])
```

Output :

```
company> db.employees.aggregate([
|   {
|     $group: {
|       _id: "$department",
|       averageExperience: { $avg: "$experience" }
|     }
|   },
|   {
|     $sort: {
|       _id: 1
|     }
|   }
| ])
[
  { _id: 'Engineering', averageExperience: 5 },
  { _id: 'Finance', averageExperience: 2 },
  { _id: 'HR', averageExperience: 4 }
]
```

6.3 Top Engineering Employees
Command :

```
db.employees.aggregate([
  {
    $match: {
      department: "Engineering"
    }
  },
  {
    $project: {
      _id: 0,
      name: 1,
      experience: 1
    }
  },
  {
    $sort: {
      experience: -1
    }
  },
  {
    $limit: 2
  }
])
```

Output :

```
company> db.employees.aggregate([
|   {
|     $match: {
|       department: "Engineering"
|     }
|   },
|   {
|     $project: {
|       _id: 0,
|       name: 1,
|       experience: 1
|     }
|   },
|   {
|     $sort: {
|       experience: -1
|     }
|   },
|   {
|     $limit: 2
|   }
[| ])
[ { name: 'David', experience: 6 }, { name: 'Robert', experience: 5 } ]
```